

1 My solutions for exercises

I'm practicing proof writing as well as problem solving here, so the writing style can vary from one solution to another.

1.1 Exercise 2.2.3

1.1.1 (a) $a \geq a$.

Proof. We seek an m such that $a = a + m$. Clearly 0 is such an m , and hence $a \geq a$.

1.1.2 (b) If $a \geq b$ and $b \geq c$, then $a \geq c$.

Proof. Let m and n be natural numbers such that $a = b + m$ and $b = c + n$. Then we have $a = (c + n) + m$. By the associative law for addition, $a = c + (n + m)$. Therefore $a \geq c$.

1.1.3 (c) If $a \geq b$ and $b \geq a$, then $a = b$.

Proof. Let m and n be natural numbers such that $a = b + m$ and $b = a + n$. Then we have $a = (a + n) + m$ and $a = a + (n + m)$, and so $n + m$ must be zero. Thus $n = m = 0$ by Corollary 2.2.9. Substituting 0 for m into $a = b + m$, we conclude that $a = b$.

1. Lemma. If $a + m = a$ then $m = 0$.

Proof. $a + m = a = a + 0$. Using the cancellation law gives $m = 0$.

1.1.4 (d) $a \geq b$ iff $a + c \geq b + c$.

Proof.

1. (i) If $a \geq b$ then $a + c \geq b + c$.

Let $a = b + m$ for some natural number m . Then $a + c = (b + c) + m$, hence $a + c \geq b + c$.

2. (ii) If $a + c \geq b + c$ then $a \geq b$.

Since $a + c \geq b + c$, there exists a natural number m such that $a + c = (b + c) + m$. Then $a + c = (b + m) + c$. By the cancellation law $a = b + m$. Hence $a \geq b$.

1.1.5 (e) $a < b$ iff $a++ \leq b$.

Proof. First, suppose $a < b$. Then there exists a natural number m such that

$$b = a + m$$

Since $a < b$ implies $a \neq b$, $m \neq 0$. Thus by Lemma 2.2.10, there is a natural number n such that $n++ = m$. Hence

$$b = a + (n++) = (a + n)++ = (a++) + n$$

Therefore $a++ \leq b$.

Conversely, suppose that $a++ \leq b$. Let m be a natural number such that $b = (a++) + m$. Then

$$b = (a++) + m = (a + m)++ = a + (m++)$$

Suppose, for contradiction, that $a = b$. Then

$$a = a + (m++)$$

and cancelling a gives $m++ = 0$. But that contradicts Axiom 2.3, which says $m++ \neq 0$. Thus $a \neq b$.

Since $b = a + (m++)$, we conclude that $a < b$.

1.1.6 (f) $a < b$ iff $b = a + d$ for some positive number d .

Proof. First, assume that $a < b$. Then $b = a + d$ for some natural number d . Suppose, for contradiction, that $d = 0$. then $b = a + 0 = a$, contradicting the fact that $a \neq b$. Hence $d \neq 0$, and we conclude d is positive.

Conversely, assume that $b = a + d$ for some positive number d . If $a = b$, $d = 0$ by the same argument used in (e), contradicting the fact that $d \neq 0$. Thus $a \neq b$ and we conclude that $a < b$.

1.2 Exercise 2.2.4

Proposition 2.2.13. Let a and b be natural numbers. Then exactly one of the following statements is true: $a < b$, $a = b$, $a > b$.

Proof. We first prove that at most one of the statements is true. If $a = b$, by definition $a < b$ and $a > b$ is false. If $a < b$ or $a > b$, then by definition $a \neq b$. Finally, if $a < b$ and $a > b$, then $a = b$ by Proposition 2.2.12.(c), a contradiction. Hence at most one of the statements is true.

We now prove that at least one of the statements is true. We proceed by induction on a . Consider the base case $a = 0$. Since $b = 0 + b$, we have $0 \leq b$ for all b . Thus either $0 < b$ or $0 = b$, and the base case is done. Now suppose inductively that $a < b \vee a = b \vee a > b$. We need to show that the proposition still holds on $a++$. If $a = b$, we have $a++ = b++ = b + 1$. Since $a++ \neq b$, $a++ > b$. If $a < b$, $a++ \leq b$ by Proposition 2.2.12.(e). Then either $a++ < b$ or $a++ = b$, and in either case we are done. Finally, if $a > b$ then for some natural number m , $a = b + m$. Now $a++ = (b + m)++ = b + (m++)$. Hence $a++ > b$. This closes the induction.

1.3 Exercise 2.2.5.

Proposition 2.2.14 (Strong principle of induction). Let m_0 be a natural number, and let $P(m)$ be a property pertaining to an arbitrary natural number m . Suppose that for each $m \geq m_0$, we have the following implication: If $P(m')$ is true for all natural numbers $m_0 \leq m' < m$, then $P(m)$ is also true. (In particular, this means that $P(m_0)$ is true, since in this case the hypothesis is vacuous.) Then $P(m)$ is true for all natural numbers $m \geq m_0$.

The solution became unnecessarily long here because I tried to give a different proof for each lemma.

1.3.1 Lemma a. $a \neq 0 \rightarrow \exists m(N(m) \wedge a = 1 + m)$.

Proof. Since a is positive, by Lemma 2.2.10 there exists a natural number m such that $a = m++$. Then we have $a = m++ = 0 + m++ = (0 + m)++ = (0++) + m = 1 + m$.

1.3.2 Lemma b. $m \neq 0 \wedge n \neq 0 \rightarrow m + n \neq 1$.

Proof 1. Suppose $m + n = 1$. by Lemma 2.2.10, there exist natural numbers p and q such that $p++ = m$ and $q++ = n$. then $m + n = (p++) + (q++) = ((p + q)++) = 1 = 0++$. by Axiom 2.4. we infer $(p + q)++ = 0$, contradicting axiom 2.3., i.e., $(p + q) \neq 0$. hence $m + n \neq 1$.

Proof 2. Suppose $m + n = 1$. by Lemma 2.2.10, there exists a natural number p such that $p++ = m$. then $m + n = (p++) + n = (p + n)++ = 1 = 0++$. by axiom 2.4 we infer $p + n = 0$. but since n is positive, $p + n \neq 0$. By Proposition 2.2.8, a contradiction. Hence $m + n \neq 1$.

1.3.3 Lemma c. $a < b++ \wedge a \neq b \rightarrow a < b$.

Proof 1 (direct). By Proposition 2.2.12.(f), $b++ = a + n$ for some positive number n . Since n is positive, by Lemma a there exists an m such that $n = m + 1$. Now $b++ = b + 1 = a + n = a + (m + 1) = (a + m) + 1$. Cancelling 1 from both sides yields $b = a + m$. Thus $b \geq a$. Since we also have $a \neq b$, we conclude that $b > a$.

Proof 2 (indirect). Suppose $a < b$ is false. Then by Proposition 2.2.13, $a > b$ must hold. Now choose two positive numbers n and m such that $b++ = a + n$ and $a = b + m$. Now $b++ = b + 1 = a + n = (b + m) + n = b + (m + n)$. Cancelling b from both sides gives $m + n = 1$. But this contradicts Lemma b, since m and n are both positive. Hence $b > a$.

1.3.4 Proof

Proof. Let $Q(n)$ be the property that $P(m)$ is true for all $m_0 \leq m < n$. We prove by induction that $Q(n)$ is true for every natural number n .

If $n = 0$ then $Q(0)$ is vacuously true, since there's no m such that $m < 0$.

Now suppose inductively that $Q(n)$. Let m satisfy $m_0 \leq m < n++$. We must show $P(m)$. We can think of two cases here: $m = n$ and $m \neq n$.

First, consider the case $m = n$. Notice that by the inductive hypothesis, $P(m')$ for all $m_0 \leq m' < n$. It follows that $P(n)$, i.e., $P(m)$, from the hypothesis of the theorem, and we are done.

Now if $m \neq n$, by Lemma c $m < n$. Thus by the inductive hypothesis we infer that $P(m)$.

This closes the induction, and we have proven that $Q(n)$ holds for every natural number n .

Using this result, we now show the conclusion of the theorem, that is, $P(m)$ for all $m \geq m_0$. For every natural number $m \geq m_0$, $Q(m++)$ holds, since $m++$ is a natural number. Together with the fact that $m_0 \leq m < m++$, we conclude that $P(m)$. This demonstrates the strong principle of induction.

1.4 Exercise 2.2.6.

Let n be a natural number, and let $P(m)$ be a property pertaining to the natural numbers such that whenever $P(m++)$ is true, then $P(m)$ is true. Suppose that $P(n)$ is also true. Prove that $P(m)$ is true for all natural numbers $m \leq n$; this is known as the principle of backwards induction.

1.4.1 Initial solution

1. Lemma. If $m \leq 0$ then $m = 0$.

Proof. Let $m \leq 0$. By definition $0 = m + n$ for some natural number n . But by Corollary 2.2.9, m and n must both be zero. Hence $m = 0$.

2. Solution

We induct on n . We first prove the principle holds on $n = 0$. Let m be a natural number such that $m \leq 0$. Then m must be 0, and from $P(0)$ we immediately obtain that $P(m)$.

Now suppose inductively that the principle holds on n . We have to prove the principle on $n++$. We first obtain the conclusion of the inductive hypothesis. We have $P(n++)$, so we infer $P(n)$ from the hypothesis. we have the requirements of backwards induction on n , and thus $P(m)$ is true for every natural number $m \leq n$.

Now let m be a natural number such that $m \leq n++$. If $m = n++$, then $P(n++)$, i.e., $P(m)$ and we are done. Otherwise, $m \leq n$ and from the earlier result we obtain $P(m)$. This closes the induction, and we have demonstrated the principle of backwards induction.

1.4.2 Solution with revised writing & structure

Using the same lemma as above.

Let $Q(n)$ be the property that: "if $P(n)$ is true, then $P(m)$ is true for all $m \leq n$ ". We prove $Q(n)$ for every natural number n , using induction.

Consider the base case $n = 0$. Let $m \leq 0$. Then m must be 0 by the above lemma, and since $P(0)$, we conclude that $P(m)$.

Suppose inductively that $Q(n)$. We have to prove $Q(n++)$. From $P(n++)$ and the definition of P , we have $P(n)$. Thus by the inductive hypothesis, $P(m)$ for all $m \leq n$.

Now let m be a natural number such that $m \leq n++$. If $m = n++$, then we substitute m for $n++$ in $P(n++)$, and we are done. Otherwise, $m \leq n$ and thus $P(m)$. This closes the induction.

1.5 The product of two natural numbers is a natural number.

Proof. Let n, m be natural numbers. We induct on n , keeping m fixed.

The base case is obvious, since $0 \cdot m = 0$, and 0 is a natural number.

Now suppose inductively that $n \cdot m$ is a natural number. Then $(n++) \cdot m = (n \cdot m) + n$ by Definition 2.3.1. Since $n \cdot m$ and m are natural numbers, $(n \cdot m) + n$ is so by the property of addition. This closes the induction.

1.6 Exercise 2.3.1.

Lemma 2.3.2 (Multiplication is commutative). Let n, m be natural numbers. Then $n \cdot m = m \cdot n$.

1.6.1 Lemma. $n \cdot 0 = 0$.

Proof. We prove by induction. The base case $0 \cdot 0 = 0$ is straightforward from the definition. Now if $n \cdot 0 = 0$, then $(n++) \cdot 0 = (n \cdot 0) + 0 = 0 + 0 = 0$.

1.6.2 Lemma. $m \cdot (n++) = (m \cdot n) + m$.

Proof. We use induction on m . For the base case, by definition of multiplication we have $0 \cdot (n++) = 0 = 0 + 0 = (0 \cdot n) + 0$, and we are done.

Now suppose inductively that $m \cdot (n++) = (m \cdot n) + m$. Then by definition of multiplication

$$(m++) \cdot (n++) = (m \cdot (n++)) + n++ = (m \cdot n + m) + n++$$

And the properties of addition gives

$$(m \cdot n + m) + n++ = m \cdot n + (m + n)++ = m \cdot n + n + (m++)$$

Finally, applying the definition of multiplication we obtain

$$(m++) \cdot n + m++.$$

Thus $(m++) \cdot (n++) = (m++) \cdot n + m++$. This closes the induction.

1.6.3 Solution

We induct on n . Consider the base case $n = 0$. Then by definition of multiplication and the first lemma, $0 \cdot m = 0 = m \cdot 0$.

Now we prove the inductive step $(n++) \cdot m = m \cdot (n++)$. We start from the inductive hypothesis $n \cdot m = m \cdot n$. We add m to both sides and obtain $(n \cdot m) + m = (m \cdot n) + m$. By definition, LHS is equal to $(n++) \cdot m$. Also, the above lemma says that $(m \cdot n) + m$ is equal to $m \cdot (n++)$. Thus we conclude that $(n++) \cdot m = m \cdot (n++)$, and this closes the induction.

1.7 Exercise 2.3.2

Lemma 2.3.3 (Positive natural numbers have no zero divisors). Let n, m be natural numbers. Then $n \cdot m = 0$ if and only if at least one of n, m is equal to zero. In particular, if n and m are both positive, then nm is also positive.

Proof. First, suppose that $n = 0 \vee m = 0$. If $n = 0$, then by Definition 2.3.1 $n \cdot m = 0 \cdot m = 0$. Otherwise, $m = 0$, and $m \cdot n = 0$ by the same argument. Since multiplication is commutative, that leads to $n \cdot m = 0$.

Conversely, suppose that $nm = 0$, and for contradiction both n and m are positive. Then by Lemma 2.2.10, $n = p++$ for some natural p . Now $n \cdot m = (p++) \cdot m$, and by definition of multiplication $n \cdot m = p \cdot m + m$. Since $n \cdot m = 0$, we have $p \cdot m + m = 0$. By Corollary 2.2.9 it follows that $m = 0$, a contradiction. Thus at least one of n and m must be zero.

1.8 Proposition 2.3.4

Proposition 2.3.4 (Distributive Law). For any natural numbers a, b, c , we have $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.

Proof. Since multiplication is commutative, we only have to show the first identity. We induct on a , keeping b and c fixed.

For the base case $a = 0$, we have the following identities:

$$0(b + c) = 0 = 0 + 0 = 0b + 0c$$

Hence the base case is done.

Now suppose inductively that $a(b + c) = ab + ac$. Then

$$\begin{aligned} (a++) (b + c) &= a(b + c) + b + c && \text{Definition of multiplication} \\ &= (ab + ac) + b + c && \text{Inductive hypothesis} \\ &= (ab + b) + (ac + c) && \text{Commutativity and associativity of addition} \\ &= (a++)b + (a++)c && \text{Definition of multiplication} \end{aligned}$$

This proves the inductive step, and thus $a(b + c) = ab + ac$ for all natural numbers a, b, c .

1.9 Exercise 2.3.3

Proposition 2.3.5 (Multiplication is associative). For any natural numbers a, b, c , we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

Proof. We use induction on a . The base case $a = 0$ is straightforward: both $(0 \cdot b) \cdot c$ and $0 \cdot (b \cdot c)$ are equal to 0 by the definition of multiplication.

Now suppose inductively that $(a \cdot b) \cdot c = a \cdot (b \cdot c)$. We need to show that $((a++) \cdot b) \cdot c = (a++) \cdot (b \cdot c)$. The left-hand side is $(a \cdot b + b) \cdot c$ by definition, which is by distributive law equal to $(a \cdot b) \cdot c + b \cdot c$. The right-hand side is $a \cdot (b \cdot c) + b \cdot c$. But by the inductive hypothesis $(a \cdot b) \cdot c = a \cdot (b \cdot c)$, so we conclude that both sides are equal. This closes the induction.

1.10 Proposition 2.3.6

Proposition 2.3.6 (Multiplication preserves order). If a, b are natural numbers such that $a < b$, and c is positive, then $ac < bc$.

Proof. By Proposition 2.2.12.(f), there exists a positive number m such that $b = a + m$. Then $bc = (a + m)c = ac + mc$. Since m and c are both positive, $mc > 0$ by Lemma 2.3.3. Thus, using Proposition 2.2.12.(f) again, we can conclude that $bc > ac$.

1.11 Corollary 2.3.7

Corollary 2.3.7 (Cancellation law). Let a, b, c be natural numbers such that $ac = bc$ and c is non-zero. Then $a = b$.

Proof. We proceed by contradiction. Suppose $a \neq b$. Then by the trichotomy of order, either $a > b$ or $a < b$. If $a > b$, by Proposition 2.3.6 $ac > bc$, which contradicts $ac = bc$. We can also obtain the same result when $a < b$. Hence $a = b$.

1.12 Exercise 2.3.4

Prove the identity $(a + b)^2 = a^2 + 2ab + b^2$ for all natural numbers a, b .

Lemma. $a^1 = a$ for all natural number a .

Proof. $a^1 = a^{0++} = a^0 \cdot a = 1 \cdot a = a$.

We now prove the identity $(a + b)^2 = a^2 + 2ab + b^2$ by the following identities:

$$\begin{aligned}
(a+b)^2 &= (a+b)^1 \cdot (a+b) && \text{Definition of exponentiation} \\
&= (a+b) \cdot (a+b) && \text{Lemma} \\
&= (a+b) \cdot a + (a+b) \cdot b && \text{Distributive law} \\
&= aa + ba + ab + bb && \text{Distributive law again} \\
&= aa + ab + ab + bb && \text{Commutativity of multiplication} \\
&= aa + 1ab + ab + bb && \text{Identity of multiplication} \\
&= aa + 2ab + bb && \text{Definition of multiplication} \\
&= a^1a + 2ab + b^1b && \text{Lemma} \\
&= a^2 + 2ab + b^2 && \text{Definition of exponentiation}
\end{aligned}$$

(Since the exercise comes immediately after the definition of exponentiation, I decided to be verbose here.)

1.13 Exercise 2.3.5

Proposition 2.3.9 (Euclidean algorithm). Let n be a natural number, and let q be a positive number. Then there exist natural numbers m, r such that $0 \leq r < q$ and $n = mq + r$.

Proof. We induct on n , keeping q fixed. Consider the base case $n = 0$. Then we can choose $m = 0$ and $r = 0$, since $0 < q$ and $0 = 0q + 0$.

Now suppose inductively that there exist natural numbers m and r satisfying the conditions. We have to find m' and r' such that $0 \leq r' < q$ and $n++ = m'q + r'$.

Since $n = mq + r$, we have $n++ = (mq + r)++ = m' + (r++)$.

By the inductive hypothesis $r < q$, and thus $r++ \leq q$. So we have two cases here: $r++ < q$ or $r++ = q$.

If $r++ < q$, then $m' = m$ and $r' = r++$, so we are done. Otherwise $r++ = q$, and then $mq + (r++) = mq + q = (m+1)q$. Hence we can choose $m' = m+1$ and $r = 0$. This closes the induction.

1.14 Exercise 3.1.1.

Show that the definition of equality in Definition 3.1.4 is reflexive, symmetric, and transitive.

Proof. Let A, B, C be sets. First, for any object x we have $x \in A \leftrightarrow x \in A$, since it's a tautology. Hence $A = A$. If $A = B$ then for any object x , $x \in A \leftrightarrow x \in B$. But that is equivalent to $x \in B \leftrightarrow x \in A$, which leads

to $B = A$. Finally, suppose that $A = B$ and $B = C$. Then for any object x , $x \in A \leftrightarrow x \in B$ and $x \in B \leftrightarrow x \in C$. Thus we have $x \in A \leftrightarrow x \in C$, which is equivalent to $A = C$.

1.15 Exercise 3.1.11.

Show that the axiom of replacement implies the axiom of specification.

Proof. Let A be a set, and let $P(x)$ be a property pertaining to the elements of A . We have to show that there exists a set X such that for any object x , $x \in X$ if and only if $x \in A$ and $P(x)$.

Let $Q(x, y)$ be the statement " $x = y$ and $P(x)$ ".

We claim that for each $x \in A$, there is at most one y for which $Q(x, y)$ is true. Let $x \in A$, and let y_1, y_2 satisfy $Q(x, y_1)$ and $Q(x, y_2)$. Then $x = y_1$ and $x = y_2$, which then implies $y_1 = y_2$. Hence our claim is true, and by Axiom of replacement there exists a set Y such that for any object z , $z \in Y$ if and only if $Q(x, z)$ for some $x \in A$.

We now prove that Y is the X we are seeking, i.e., for every z , $z \in Y$ if and only if $z \in A$ and $P(z)$.

First, suppose $z \in A$ and $P(z)$. To show that $z \in Y$, we have to find an $x \in A$ such that $Q(x, z)$ holds. Indeed, $Q(z, z)$ holds, since $z = z$ and $P(z)$. Thus, z is such an x , and therefore $z \in Y$.

Conversely, let $z \in Y$. Then there exists an $x \in A$ such that $Q(x, z)$, i.e., $x = z$ and $P(x)$. Since $x = z$, it follows that $P(z)$ and $z \in A$.

Hence, Y is the set we were looking for, and this completes our proof.